LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Principle of operation of gas turbine engine
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Principle of operation of gas turbine engine.
5. Specific Objectives	9.1.1.2 Trainee will be able to explain the principle of operation of gas turbine engine.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of engine cutaway model; Safety: Do not touch hot components; Supervision: Instructor monitors model handling
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a gas turbine engine?
(b) Name the four main components of a gas turbine engine.
(c) How does a gas turbine engine differ from a piston engine?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic components of gas turbine engine • Air intake and compression process • Combustion chamber operation • Turbine expansion and work extraction • Exhaust jet propulsion	L & Ds	PPT, IB	15 min
(b)	• Four stages of gas turbine cycle • Energy conversion in compressor and turbine • Role of combustion in thrust generation • Mass flow continuity through engine • Relationship between pressure and temperature	L & Ds	PPT, PDF	20 min
(c)	• Newton's third law in jet propulsion • Thrust vector and engine orientation • Efficiency factors in gas turbine operation • Impact of airspeed on engine performance • Basic limitations of gas turbine principle	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the gas turbine engine converts fuel energy into thrust.
(b) Describe the role of the turbine in the gas turbine cycle.
(c) Identify one key difference between gas turbine and piston engine operation.

12. Home Assignment and Follow Up	Prepare a brief note on how the principle of jet propulsion applies to the next lesson on turbojet and bypass engines.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to propulsion, principle of jet propulsion, method of jet engine, bypass engine, single and multi-spool engine, turbo jet, turbo fan, turbo prop, turbo shaft, turbo rocket engines etc.	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min